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23R-D25 (MAT5104C)

2025

MATHEMATICS

(MAJOR - I)

Paper : MAT5104C

**(Theory of Real Functions and Riemann
Integration)**

Full Marks : 60

Time : 2½ hours

***The figures in the margin indicate
full marks for the questions.***

1. Answer the following questions : $1 \times 7 = 7$

(a) Define cluster point of a set.

(b) "A function which is continuous in a closed interval is also uniformly continuous in that interval"—Write true or false.

(c) If a function f is increasing on an interval I , what conclusion can be drawn about the sign of the derivative $f'(x)$ on that interval?

(d) If a function is differentiable at x_0 then what conclusion can be drawn about continuity of f at that point?

(e) Evaluate $\lim_{x \rightarrow \infty} \left[\frac{e^x}{x^3} \right]$ by using

L' Hospital's rule.

(f) State the first fundamental theorem of calculus.

(g) Write the definition of beta function $\beta(m, n)$. $-\int_0^{\infty} x^{m-1} (1-x)^{n-1} dx$

2. Answer all parts of the questions: $2 \times 4 = 8$

(a) Is the function $f(x) = |x|$ is continuous at $x = 0$? Justify your answer.

(b) Verify Rolle's theorem for $f(x) = x^2 - 4x + 3$ on $[1, 3]$.

(c) Find Maclaurin's series expansion of $f(x) = \sin(x)$ up to the term containing x^3 .

(d) Test the convergence of the improper integral $\int_1^{\infty} \frac{1}{x^p} dx$.

3. Answer (a), **one** from (b) or (c), and **one** from (d) or (e): 5×3=15

(a) Let $f(x)$ and $g(x)$ be two differentiable functions on I containing the point a . Suppose that

$$\lim_{x \rightarrow a} f(x) = 0 \text{ and } \lim_{x \rightarrow a} g(x) = 0 \text{ with}$$

$g'(x) \neq 0$ for all $x \in I, x \neq a$ and the

$\lim_{x \rightarrow a} \left[\frac{f'(x)}{g'(x)} \right]$ exists then show that

$$\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] \equiv \lim_{x \rightarrow a} \left[\frac{f'(x)}{g'(x)} \right].$$

(b) Define uniform continuity and show that $f(x) = \frac{1}{x}$ is not uniformly continuous on $(0, 1)$.

OR

(c) Define an increasing and a decreasing function. Determine the intervals of increase and decrease of the function

$$f(x) = x^3 - 3x^2 + 2.$$

- (d) Expand $f(x) = e^x$ about $x = 0$ upto the term containing x^4 , using Taylor's theorem with Lagrange's remainder.

OR

- (e) Define Gamma function and show that $\Gamma(n+1) = n\Gamma(n)$.

4. Answer (a) or (b):

10

- (a) State and prove squeeze theorem and use it to show that

~~$\lim_{x \rightarrow 0} f(x) = L$~~
 ~~$\lim_{x \rightarrow 0} g(x) = L$~~
 ~~$\lim_{x \rightarrow 0} h(x) = L$~~

$$\lim_{x \rightarrow 0} \left[x \sin \frac{1}{x} \right] = 0.$$

OR

- (b) Let $f: I \rightarrow \mathbb{R}$ be continuous on I and let $a, b \in I$. If $k \in \mathbb{R}$ be such that $f(a) < k < f(b)$ then prove that there exists a point $c \in I$ and $a < c < b$ such that $f(c) = k$. Use this prove to show that $f(x) = x^3 - x - 2$ has a zero in the interval $[1, 2]$.

5. Answer (a) **or** (b):

10

(a) State and prove Caratheodory's theorem and use it to find derivative of $f(x) = x^3$.

OR

✓ (b) Let $f: A \rightarrow \mathbb{R}$ and c be a cluster point of A . If $\lim_{x \rightarrow c} f(x) = L$, then prove that for every sequence (x_n) in A that converges to c , the sequence $(f(x_n))$ converges to L . Using this criterion show that $\lim_{x \rightarrow 0} \operatorname{sgn}(x)$ does not exist.

6. Answer (a) **or** (b):

10

✓ (a) Let f be a bounded real-valued function on $[a, b]$ then define the upper sum $U(p, f)$ and the lower sum $L(p, f)$ corresponding to a partition p of $[a, b]$. Prove that $L(p, f) \leq U(p, f)$ and if p^* is a refinement of the partition p then $L(p, f) \leq L(p^*, f)$.

OR

(b) If a function f is continuous on $[a, b]$ then prove that there exists a $c \in [a, b]$

such that $\int_a^b f(x) dx = (b - a)f(c)$.

Verify the proof if $f(x) = x^2$ on $[0, 2]$.
